Chapter 13, Problem 89P

Problem

A 240/120-V rms power transformer is rated at 10 kVA. Determine the turns ratio, the primary current, and the secondary current.

Step-by-step solution

Step 1 of 1

$$n = \frac{V_2}{V_1}$$

$$= \frac{120}{240}$$

$$= 0.5$$

$$S = I_1 V_1$$

Or,

$$I_1 = \frac{S}{V_1}$$

$$= \frac{10\times10^3}{240} = 41.67\ A$$

$$S = I_2 V_2$$

Or,

$$I_2 = \frac{S}{V_2}$$

$$= \frac{10^4}{120} = 83.33\ A\ .$$